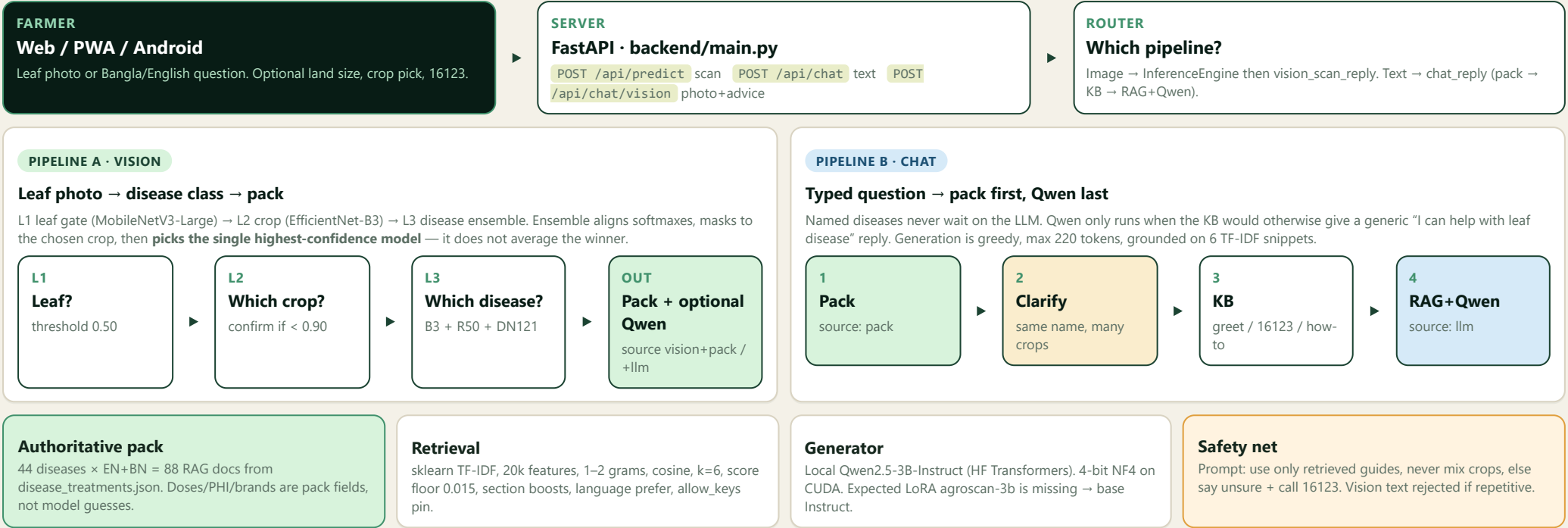


How everything works — vision, pack, RAG, and Qwen

Two farmer entries, one FastAPI server. A leaf photo goes through three vision levels, then a verified disease pack (Qwen only writes a short wrap). A typed question hits the pack first, then the knowledge base, and only leftover open questions go through TF-IDF retrieval into local **Qwen2.5-3B-Instruct**.



Vision scan — three levels, then farmer wording



How the “best answer” is chosen

- Each disease model emits softmax, aligned onto the shared class list (older 38-class checkpoints do not dilute Bangladesh-only classes).
- Probabilities are **masked to the L2 crop** so a tomato photo cannot win as rice blast.
- Code comment is explicit: *NEVER average — pick the single model with the highest confidence.*
- Uncertain if confidence < 0.45; UI warns if < 0.80 (retake photo / call an officer).
- Class label looks like **Rice__Blast** → plant + condition + pack lookup via advice_for().

If leaf_type.pt is missing, crop is estimated by summing disease-class probabilities by plant. L1 missing means the gate is skipped (is_leaf = true).

After the class is known

Header always

Leaf % · crop · disease · confidence. Bangla or English.

Try Qwen wrap · generate_vision_feedback

Does *not* call retrieve(). Prompt = scan header + pack title, 5 steps, 4 treatments, 3 prevention lines. 3–6 short sentences. Max 180 new tokens, greedy.

Quality gate · _llm_text_is_bad

Reject if one word is ≥22% of the text, or 6 identical tokens in a row. Known 3B collapse mode.

Fallback

vision+llm if Qwen is clean · vision+pack if rejected or unloaded · bare vision + “call 16123” if no pack.

Status returned to UI	When	source field
not_leaf	L1 says this is not a leaf	vision
need_crop	L2 confidence below 0.90 and farmer has not picked	vision
no_disease	No class from L3	vision
ok	Class + header + Qwen or pack body	vision+llm / vision+pack / vision

Chat router — pack is the product, Qwen is overflow

1. resolve_disease_query(message)

Score pack/KB candidates. If one disease+crop is clear → format the full pack reply (symptoms, steps, chemicals, 16123). **source: pack. Qwen is not called.**

2. Ambiguous name → clarify

Same kb_key on several crops (e.g. blight). Return options; wait for confirm_class. **source: clarify**

3. chatbot_reply / structural KB

Greetings, emergency, government links, “how do I use AgroScan”, scan-context follow-ups. If the KB text is already specific, return it. **source: kb / context / pack**

4. Only then generate_reply — RAG + Qwen

Runs if the KB answer is generic (“I can help with plant-leaf...”). retrieve(k=6) → prompt → Qwen. Fail → KB fallback. **source: llm**

WHY THIS ORDER

Chemical doses are a legal/safety surface. The pack is DAE/PPW-backed text. The 3B model is instructed never to invent products. Routing enforces that: named diseases never reach generation.

What a pack answer contains

- Crop + disease title (EN and BN fields)
- Pathogen, season, severity, source, last verified
- Cultural control and fertilizer (BARC FRG-style)
- Chemical options: product, AI, dose/L, timing, interval, max sprays, PHI, BD brands
- Registration: verified AP, or “confirm with dealer/PPW”
- Organic options, safety, legal, when to call 16123

Prompt Qwen actually sees (open questions only)

```
### Instruction:
You are AgroScan... ONLY use the retrieved guides below. Never mix one crop/disease into another. If guides do not cover the question, say you are unsure and suggest calling 16123. Do not invent products, doses, or symptoms.
Matched disease record (authoritative): {title + summary}    ← if named match
Retrieved guides:
[1] ... [6] section chunks    or “none. Say you are unsure.”
### Input:
{farmer message}
### Response:
```

Bangla system prompt is used when lang starts with bn. allow_keys pins retrieval to the matched class_name / kb_key so maize blight steps cannot leak into potato late blight. Prompt truncated at 2048 tokens; generation max 220, do_sample=False.

RAG pipeline — lexical TF-IDF over the Bangladesh pack

SOURCE OF TRUTH

disease_treatments.json

DAE / PPW / BRRI / BARC-backed pack. Builder does not invent doses or AP numbers.

BUILD_ALL()

44 × EN+BN = 88 docs

Healthy classes skipped. Each disease: overview, symptoms, prevention, treatment, process.

CHUNKS

Section documents

Aliases (title, crop, BN names, kb_key, Class__Disease) prepended for keyword hit.

INDEX

In-RAM sklearn

TfidfVectorizer 20k features, ngram (1,2). @lru_cache. No Chroma/FAISS.

retrieve(query, lang, k=6)

Step	Rule
Query	optional extra (scan class) + user text
Score	cosine similarity on TF-IDF vectors
Boost	+0.15 prevention/treatment; +0.12 process/symptoms if keywords match (EN+BN)
Language	prefer bn chunks if lang is bn, then fill from the rest
Floor	drop scores < 0.015
Pin	PlantVillage class_name with ____ must match exactly; else fuzzy kb_key
Dedupe	(lang, key, section)
Length	treatment/process 2200 chars; others 1600

What a treatment chunk looks like

Rice — Rice blast example (pack, not model): Tricyclazole 75% WP at 0.6 g/L, spray at booting and 50% heading, max 2 sprays, PHI 21 days, brands Trooper/Beam/Blasticure. Rotate to Nativio 75 WG. Stop urea; add MoP. Neck blast cannot be cured once the panicle is infected. Pack notes: AP numbers often “confirm with dealer” — blast vs brown spot have **opposite nitrogen advice**; retrieval pinning exists so those two are not merged.

This is not neural RAG

No dense embeddings, no Qwen-as-retriever, no vector DB. “Diamond spots on rice” will rank blast if those words sit in the chunk. A paraphrase with no shared n-grams may miss. Legacy DISEASE_GUIDES are indexed alongside the pack when the pack exists.

Qwen’s job in RAG

Generation only. It rewrites the retrieved sections into a farmer reply. It does not search. If snippets are empty, the prompt orders it to say it is unsure.

CPU vs GPU

Retrieval is CPU sklearn and always on. Qwen needs CUDA in auto mode (AGROSCAN_LLM_ENABLED=auto). Docker image is CPU PyTorch — LLM likely stays off there.

Rebuild

python -m agrovet.rag_build writes rag_disease_docs.json. clear_index_cache() drops the in-memory TF-IDF matrix.

Qwen serving vs training — and how results actually land

What is loaded at chat time

Knob	Value in code
Base	Qwen/Qwen2.5-3B-Instruct
Adapter expected	llm_data/adapters/agroscan-3b — not on disk
Adapter present	agroscan-1.5b — refused on the 3B base
Quant	4-bit NF4, else fp16 CUDA, else CPU fp32
Decode	greedy, 220 / 180 tokens, stop on ###
Enable	auto = CUDA available; pagefile error 1455 disables for the process
Preload	off unless AGROSCAN_LLM_PRELOAD=1

What was trained (not served)

Item	Count / setting
Train JSONL	69,238
Eval JSONL	2,000 — no logged eval metrics in repo
krishokchat	58,299
crop CSVs	~10k
AgroScan disease EN+BN	94 + 55
Train model	Qwen2.5-1.5B-Instruct, LoRA r=16

The fine-tune and the serve model are different sizes. Live RAG answers are stock 3B Instruct + retrieved pack text.

What the farmer actually gets today

Path	Typical question	Engine	Quality in practice
Named disease	“ধানের ব্লাস্টে কী স্প্রে?”	Pack formatter	Strong. This is the designed happy path. Caveat: many AP numbers still “confirm with dealer”.
Ambiguous name	“blight কী করব?”	Clarify UI	Working: asks which crop.
Scan with pack	Photo of blast leaf	Vision + pack, Qwen wrap if clean	Diagnosis from ensemble; advice from pack. Qwen wrap is fragile (repetition gate).
Open chat	Paraphrase with no disease name	TF-IDF + base 3B	Unguaranteed. No recall/faithfulness eval in repo. Keyword retrieval; small greedy model; no 3B LoRA.
Hello / 16123 / how-to	“হ্যালো”, “হেল্পলাইন”	KB	Deterministic, no LLM.

Trust this

Pack JSON + vision class → farmer treatment. Routing, allow_keys, and the “only use retrieved guides” prompt exist to keep doses inside that pack.

Do not over-claim

There is no RAG eval set. The 1.5B LoRA is unused at serve. Docker/CPU likely never loads Qwen. GGUF export is still absent.

If you raise quality next

Train/attach agroscan-3b or serve 1.5B on a 1.5B base. Add a 20-query retrieval check that blast and brown-spot nitrogen advice never swap.